

Sentinel

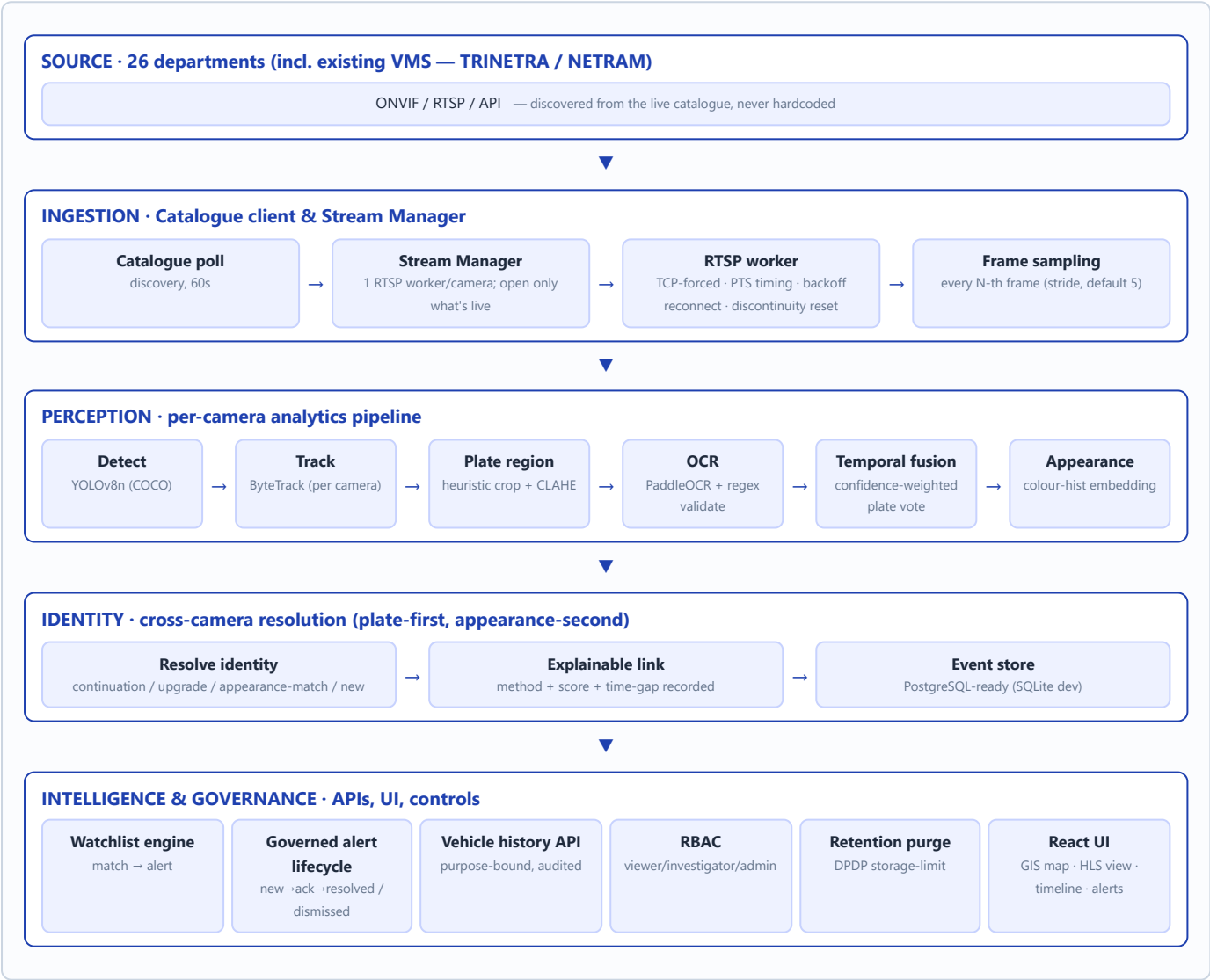
Interoperability & Intelligence Layer for Gujarat's Multi-Department CCTV Network
Gujarat Police Innovation Challenge 2026 · Full Architecture, Algorithms & Models · Generated 2026-09-03

Solution model — Hybrid (Model 1 + Model 2), Model 3/4 documented as roadmap. Model 1 (Centralised CCTV Registry & GIS Mapping, mandatory) is the common registry/GIS/governance index; Model 2 (Unified Viewing & Metadata Analytics) connects directly to each department's RTSP/ONVIF surface — no new middleware. The honest differentiator is **not** detection/ANPR (commercially commodity in India) but the **open, explainable vehicle-identity-resolution layer**: a persistent vehicle entity built from uncertain observations with an auditable link trail, degrading gracefully to appearance/attribute tracking when a plate can't be read.

REAL / VERIFIED **WORKING SUBSTITUTE** **HONEST GAP** **ROADMAP (NOT BUILT)**

Every component below is labelled with its true status. This is a government/police evaluation — visible gaps are stated rather than papered over.

1. System architecture (layered)



Core principle: centralize metadata / events / identity / GIS / governance — **not** every raw video frame. 80,000 cameras × ~2 Mbps ≈ 160 Gbps / ~1.7 PB per day makes full central video ingestion non-credible; structured events are ~3 orders of magnitude smaller (≈640 Mbps at full scale).

2. Models used

Stage	Model / technique	Provenance & licence	Status
Vehicle detection	YOLOv8n, pretrained COCO (car/truck/bus/motorcycle/person)	Ultralytics — AGPL-3.0 (project is open-source)	REAL / VERIFIED

Within-camera tracking	ByteTrack (<code>model.track(tracker="bytetrack.yaml")</code>), one persistent instance per camera	Ultralytics built-in; needs <code>lap</code>	REAL / VERIFIED
Plate OCR	PaddleOCR 2.9.1 (angle-cls, English) + regex plate-format validation	PaddlePaddle — Apache 2.0 (version-pinned)	REAL / VERIFIED
Plate localization	Heuristic crop (lower-middle third of vehicle bbox) + CLAHE enhancement	OpenCV; <code>YoloPlateDetector</code> seam ready for a fine-tuned model	WORKING SUBSTITUTE
Appearance Re-ID	Colour-histogram encoder: HSV 2-D hist (32) + 4×4 grayscale template (16), L2-normalized; cosine similarity	OpenCV/NumPy — own implementation	REAL / WORKING
Vehicle attributes	Dominant colour (median of strided sample → nearest of 10 named buckets); type from detector label; pixel dimensions	OpenCV/NumPy — own implementation	REAL / WORKING
Make / model	<code>StubMakeModelClassifier</code> — returns <code>None</code>	No credible open-source Indian-market classifier exists	HONEST GAP
Re-ID upgrade path	FastReID / OSNet (VeRi-776)	JDAI-CV — seam exists, weights not bundled	ROADMAP
Detector upgrade path	IISc UVH-26 (Indian-CCTV YOLOv11 / RT-DETRv2), OCR: Awiros ANPR (PP-OCRv5)	IISc AIM / Awiros — evaluated, needs validation / PaddleOCR 3.x	ROADMAP

Honest accuracy note: OCR is verified on clean synthetic plate text (>99% conf) and detection against real photos; **not** yet measured against real Indian-plate footage. Field accuracy comes from detector + regex validation + temporal fusion working together, not the OCR engine alone.

3. Algorithms — per stage

Stage	Algorithm	Why this choice
Discovery	Interval-pollled catalogue reconciliation; open/close workers to match live set	Camera IDs/availability change; "pace your load" — don't hold idle streams
Stream timing	RTSP forced over TCP; all timing from <code>PTS</code> (<code>CAP_PROP_POS_MSEC</code>), never wall-clock / <code>CAP_PROP_FPS</code>	Gateway replays a buffered GOP faster than real-time; PTS is the only correct clock for velocity/dwell
Resilience	Exponential-backoff reconnect (2s→30s); PTS-going-backward ⇒ scene-discontinuity ⇒ state reset	Sandbox loops recordings; a hard cut must reset tracker + detector, not produce garbage across it
Detection	YOLOv8n inference on sampled frames (stride-N), vehicle-class filter	Mobile-scale model bounds GPU cost per camera; COCO classes suffice for vehicle detection
Tracking	ByteTrack (Kalman-predicted motion + 2-stage association); IOU-overlap fallback for unconfirmed detections / stub detector	Robust cross-frame association; a library already in the dependency tree — no new model
Plate enhancement	Grayscale → CLAHE contrast normalization → cubic upscale	Dependency-free, measurably lifts OCR hit-rate on small/low-contrast crops
OCR validation	PaddleOCR read → strip non-alphanumerics → <code>fullmatch</code> Indian plate regex <code>[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}</code> ; keep highest-conf match	Regex is the Indian-plate specialization; rejects background text / partial garbage
Temporal plate fusion	Per track, sum confidence per candidate plate across frames; consensus = max vote-mass; confidence = that plate's share of total mass	A single frame's read is unreliable (glare/blur/angle); voting across the whole track is far more robust
Low-confidence suppression	Fused plate confidence < threshold (default 0.5) ⇒ treat as UNREAD; vehicle still logged anonymously	A shaky read must not spawn a false identity or false watchlist alert (Flock/Roseville lesson)
Motion attributes	Dwell = last-first PTS; speed = bbox-centre displacement ÷ elapsed PTS; direction = <code>atan2</code> of displacement — all image-plane, not calibrated km/h	Enables dwell / wrong-way / stopped-in-zone filters from data already captured; honest about not being real-world units
Appearance embedding	HSV hue-sat histogram ⊕ 4×4 grayscale template, L2-normalized	Distinguishes "white SUV" vs "red hatchback" across angles with zero extra dependencies
Cross-camera identity	Plate-first: same plate → continue; new plate + matching anonymous appearance → <i>upgrade</i> . Appearance-second: cosine ≥ 0.80 within a 15-min window → match; else new identity	Plate is the strongest signal; appearance carries identity through gaps where the plate can't be read
Explainability score	Per link: record method + cosine score + time-gap. UI fuses plate-conf 50% / appearance 35% / recency 15% — labelled a transparency aid, not a calibrated probability	Answers "why was this vehicle linked?" — a jury-facing, human-in-the-loop beat
Watchlist match	In-memory normalized-plate cache checked at event write-time; exact match → Alert row	Keeps the hot path off the DB; instant alerting
Alert lifecycle	Enforced state machine <code>new→acknowledged→resolved</code> (+ <code>dismissed</code>); illegal transition → 409; each move records who/when	Governed entity lifecycle (Palantir pattern); false-positive dismissal is a first-class auditable action
Access & governance	Ranked RBAC (viewer<investigator<admin), SHA-256-hashed keys, purpose-bound audit logging, scheduled retention purge, same-host SSRF guard on HLS proxy	Model 1's required RBAC + DPDP storage-limitation, built as real controls not a checklist

4. Technology stack

Backend & data

- **FastAPI** — REST API, RBAC-gated routes
- **SQLAlchemy** — SQLite (dev) → PostgreSQL/PostGIS-ready via `DATABASE_URL` ; idempotent additive migrations
- **OpenCV** — RTSP ingest, frame ops, enhancement, colour/embedding math
- **Ultralytics (YOLOv8 + ByteTrack)** on Torch
- **PaddleOCR 2.9.1** on PaddlePaddle 2.6.2

Frontend & delivery

- **React + Vite** — 4 tabs: Map/Registry, Vehicle Tracking, Watchlist, Registry Ops
- **Leaflet** — Model 1 GIS map (health-colour-coded)
- **hls.js** — live view via authenticated backend HLS proxy (WHEP is not wired — stated)
- All components open-source (hackathon Open-Source requirement)

5. Scalability shape & known gaps

Scale (see SCALABILITY.md)

- **Edge inference, not central video** — emit structured events (~1 KB), keep frames regional
- ~20 cameras/GPU (mobile-scale models) → ~4,000 edge units at 80k cameras
- Horizontal: shard cameras across stream-manager instances; events to a bus (Kafka — roadmap)
- Re-ID at scale: candidate-filter by time+region before vector search (pgvector/Milvus) — **ROADMAP**

Stated gaps (not hidden)

- Real Indian-plate OCR accuracy unmeasured on live footage
- Plate localization is heuristic, not a fine-tuned model
- Make/model classification not implemented
- WHEP low-latency preview not built (HLS proxy is the path)
- Department-scoped RBAC designed, not built (flat roles today)
- Sandbox RTSP validated in an earlier session; reconfirm before demo

Sentinel · Gujarat Police Innovation Challenge 2026 · Model 1 + Model 2 hybrid. Source of truth: [HLD.md](#), [SCALABILITY.md](#), [STRATEGY.md](#), [COMPETITIVE_TEARDOWN.md](#), and the [sentinel-solution/](#) backend. Every capability shown as "Real / verified" has been exercised end-to-end against the actual code, not asserted from design intent.